Maura Kelleher

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EDUCATION

Massachusetts Institute of Technology (MIT) Cambridge, MA
Candidate for Masters in Computer Science and Engineering - Concentration in Machine Learning and AI May 2026

Massachusetts Institute of Technology (MIT) Cambridge, MA
Bachelors in Computer Science and Engineering, Minor in Business Analytics May 2025

- GPA: 4.9/5.0
- Selected Coursework: Advances in Computer Vision, Quantitative Methods for Natural Language Processing, Software Construction, Machine Learning, Design and Analysis of Algorithms, Optimization Methods, Computation Structures
- Activities: Varsity Track & Field; TA for Intro to Machine Learning, Probability & Statistics, Robotic Manipulation

Los Altos High School Los Altos, CA

- GPA: 4.67/4.0 2017 - 2021

EXPERIENCE

Niantic Spatial - Augmented Reality Mapping and World Perception Team San Francisco, CA
Computer Vision Intern June 2025 - Present

- Improving the quality of 3D Gaussian Splats for 3D reconstruction by implementing novel training optimizations

MIT-IBM Watson AI Lab - IBM Research Cambridge, MA
Graduate Student Researcher January 2025 - Present

- Engineering a video-to-prosody pipeline for the automatic generation of tennis commentary from match videos

Niantic Labs - Augmented Reality Mapping and World Perception Team Bellevue, WA
Software Engineering Intern May - August 2024

- Trained novel depth estimation models on simulated and real world data for long-range reconstruction of 3D scenes
- Improved mesh rendering by conducting varied trials, experimenting with different architectures and parameters

MIT Media Lab - Fluid Interfaces Group Cambridge, MA
Undergraduate Student Researcher February - May 2024

- Collaborated on a study to investigate the effects of smartwatch rhythmic vibration on memory, sleep and cognition
- Developed a full-stack MERN app, allowing study participants to complete a customized memory task from home

BiomX - Bioinformatics Team Ness Ziona, Israel
Data Science and Machine Learning Intern May - August 2023

- Developed robust Python-SQL pipelines to facilitate retrieval and processing of data from large databases
- Designed a locally-hosted chat interface, allowing employees to analyze proprietary company data without exposing it

MIT Computer Science & Artificial Intelligence Laboratory - MIT App Inventor Cambridge, MA
Undergraduate Student Researcher September 2021- December 2023

- Built an end-to-end system that allows users to build complex apps from a natural language description
- Collaborated to conduct qualitative research to uncover patterns in individuals' attitudes towards conversational AI
- Led 40 participants in 3 days of conversational AI programming sessions and workshops in the United Arab Emirates

PROJECTS AND ACTIVITIES

MIT Global Teaching Labs Program Chile & Andorra
Teacher January 2024, 2025

- Completed 2 month-long teaching placements. Collaborated with local educators to teach Math & Computer Science
- Successfully instructed diverse classes of 20-30 high school students, providing critical one-on-one support

MIT AI Alignment Cambridge, MA (Remote)
AI Safety Fellow June 2023 - August 2023

- Reviewed literature on AGI, deception, RL reward misspecification; engaged in discussions on safe AI development

PUBLICATIONS

- J. Van Brummelen, M. C. Tian, **M. Kelleher**, N. H. Nguyen. "Learning Affects Trust: Design Recommendations and Concepts for Teaching Children---and Nearly Anyone---about Conversational Agents". (*EAAI-23)*
- J. Van Brummelen, **M. Kelleher**, M. C. Tian, N. H. Nguyen. "What Do Children and Parents Want and Perceive in Conversational Agents? Towards Transparent, Trustworthy, Democratized Agents". *(ACM IDC-23)*

SKILLS AND INTERESTS

Technical: Proficient in Python, Java, C, Javascript, HTML, React, CSS, SQL, PyTorch; VS Code, GitHub, LaTeX
Interests: Sports Analytics, Sleep & Exercise Wearables, Technical Consulting, Teaching, Wakeboarding, Running, Orcas